

1.

(a) (i) $\eta = F / 6\pi r v$

Units of $\eta = \text{kg m s}^{-2} / \text{m (m s}^{-1}) = \text{kg m}^{-1} \text{s}^{-1}$ [M1], require “unit of” or equivalent

(ii) 1.

$$F = 6\pi\eta r v = 6 (3.142) (0.13) (0.01) (2.7) = 0.06617 \text{ N} \quad \text{A1}$$

$$\frac{\Delta F}{F} = \frac{\Delta \eta}{\eta} + \frac{\Delta r}{r} + \frac{\Delta v}{v}$$

$$\frac{\Delta F}{0.06617052} = \frac{0.02}{0.13} + \frac{0.1}{2.0} + 0.05 \quad 00070777$$

$$\frac{\Delta F}{0.06617052} = 0.154 + 0.050 + 0.050 = 0.254$$

$$\Delta F = 0.02 \text{ N (1 sf)}$$

M1

A1

$$F = (0.07 \pm 0.02) \text{ N}$$

2. Measurement of viscosity, η , should be improved as it has the highest fractional uncertainty

A1

2 (a)

From Newton’s third law, if body A exerts a force on body B, then body B will

M1